AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 3 · WORKBOOK

Sensors Are Messengers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected measurement record.

Repair a record, trace a boundary, compare pairs, limit a conclusion and plan the reserved new investigation.

Keep one habit: Keep the original reading, the conditions and the reason for your conclusion.

MAIN LESSON · SECTION 1

Recall · What did the summary lose?

> Retrieve Week 2 before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from **Input**, **Process**, **Output**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

▼ **Earlier wording, alternative practice and picture transcripts**

Input, Process, Output

MAIN LESSON · SECTION 12

W1 · Repair an incomplete record

> Name missing fields without inventing observations.

Your evidence record

A notebook entry says 24 at lunchtime. Rewrite it as a useful air-temperature record by stating which missing fields must be collected. Do not invent values that were not observed.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the warning boundary

> Apply greater than 28 carefully, then separate the rule from measurement quality.

Your evidence record

A warning rule is recorded temperature greater than 28 degrees C. State its **output** for 27, 28 and 29 degrees C. Explain why an off warning does not prove that the **sensor** is working correctly.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Compare without choosing a winner

> Calculate paired differences and explain what a reference would add.

Your evidence record

At the same three times, A records 18, 19 and 18 degrees C and B records 21, 22 and 21 degrees C in the same location. Calculate the paired differences. What is established, and what further **evidence** would help judge accuracy?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit the watering conclusion

> Distinguish a temperature warning from evidence about roots and watering.

Your evidence record

An air-temperature **sensor** reports 29 degrees C and a rule turns on a warning. Someone writes The plant is thirsty.

Explain the unsupported step and name information more directly relevant to deciding whether watering is needed.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Investigate a new measurement dispute

> Make one first attempt on the sound-display case before feedback.

Read and explain independently

A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternative practice and picture transcripts**

Independent case: Investigate a new measurement dispute

MAIN LESSON · SECTION 9

Project · Choose the measurement you need

> Name a sensor only when it serves the sorter’s real task.

Your project map

- Mark which **sensor**, if any, would collect each Classroom Sorter **input**; prefer direct human entry when a **sensor** is unnecessary.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and revisit

> Keep the first attempt, its repair and a later changed-case check.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Check what your evidence shows

> Use work you can point to when judging independence.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Boundary practice copy**

REFERENCE · SECTION 2

Reference · Boundary practice copy

> This is the same W2 attempt. Use one response record.

Use this when helpful

Use W2: Trace a boundary for **Sensor**s and **measurement**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Does stability prove accuracy?

REFERENCE · SECTION 3

Extra practice · Does stability prove accuracy?

> Explain which evidence would support closeness to a reference.

Use this when helpful

Use the worked case for **Sensor**s and **measurement**. Proposed explanation: “The more stable device must be accurate”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · The light-sensor target

REFERENCE · SECTION 5

Reference · The light-sensor target

> Use the target to check what the original unknown units permit.

Use this when helpful

Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright? Design a fair comparison without using a camera or personal **data**.

▼ **Extra practice · Light and sound**

REFERENCE · SECTION 6

Extra practice · Light and sound

> Try before checking the worked explanation.

Use this when helpful

A light **sensor** reads 15 in a cupboard and 90 near a window. A microphone records a clap.

Which measures a light-related **signal**? Can either directly measure happiness?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · Readings and comfort**

REFERENCE · SECTION 7

Extra practice · Readings and comfort

> Keep this practice distinct from the reserved independent case.

Use this when helpful

A station reports light level 18 and temperature 22 degrees C. Which **sensor** produced each reading? Can either reading alone tell whether a child feels comfortable? Explain.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 3 Test and reflect

▼ **Extra practice · Design a comparison**

REFERENCE · SECTION 8

Extra practice · Design a comparison

> State a prediction before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Light-comparison response**

REFERENCE · SECTION 11

Reference · Light-comparison response

> This teaching case is practice, not a fresh assessment.

Use this when helpful

Two light **sensor**s report 40 and 80 in different rooms. Their settings and **unit**s are unknown. Can you conclude the second room is twice as bright? Design a fair comparison without using a camera or personal **data**.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

▼ Reference · Sound-display response copy

REFERENCE · SECTION 15

Reference · Sound-display response copy

> This duplicates W4. Reuse its first-attempt record.

Use this when helpful

A classroom sound display near a speaker reads higher than one near the door. A student declares the door device broken. Propose a safe comparison using normal classroom sound.

Predict one result consistent with location explaining the difference and one result that would leave a device difference worth investigating.

> **My reasoning and evidence**

▼ Extra practice · Paired light readings

REFERENCE · SECTION 18

Extra practice · Paired light readings

> Use the picture and name the support you needed.

Use this when helpful

Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.

On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Neither device has been compared with a trusted reference.

AI ACADEMY / AI-1/3

Simultaneous light readings

	Sensor A	Sensor B	B minus A
Time 1	40	45	5
Time 2	50	55	5
Time 3	60	65	5

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/3

Sensors Are Messengers

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Same shelf
Place both devices together and allow them to settle.

↓

2. Paired readings
Record three matching times: (40,45), (50,55), (60,65).

↓

3. Compare gaps
B minus A is 5, 5 and 5 units.

↓

4. Check a reference
A trusted reference is needed to decide which reading is closer.

CHANGE ONE THING
Move only B into brighter light.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why can we calculate the earlier gap but not predict an exact new B reading?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions.

▼ **Reference · Sorter portfolio record**

REFERENCE · SECTION 19

Reference · Sorter portfolio record

> Extend your main project record if needed.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: sensor pictures and feedback**

REFERENCE · SECTION 20

Extra practice: sensor pictures and feedback

> Try each question before opening its explanation. Use one paper or browser copy.

First supported question

- Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken.
- Which labelled step matches this detail? A trusted reference is needed to decide which reading is closer.
- 1. Compare gaps
- 2. Check a reference
- 3. Paired readings
- 4. Same shelf

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check a reference A trusted reference is needed to decide which reading is closer. A repeated difference of 5 supports investigating an offset between devices.

Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors. Preserve paired readings instead of adjusting them silently.

Changed-condition question

- Changed-case practice: Move only B into brighter light. Which conclusion is supported by the supplied facts?
- 1. The next gap could change for reasons other than a difference in device accuracy.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed condition**

After your attempt, compare with the supported explanation: The next gap could change for reasons other than a difference in device accuracy.

The comparison now changes location as well as device; there is no supplied reading from the new location.

Repair and save

Review **Input**, **Process**, **Output** (ai-1/2). Goal: Trace information through a system without confusing information with a physical action. Short practice: Use the richer record to find blue pencils in Box B.

Explain why total alone was insufficient.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternative practice and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

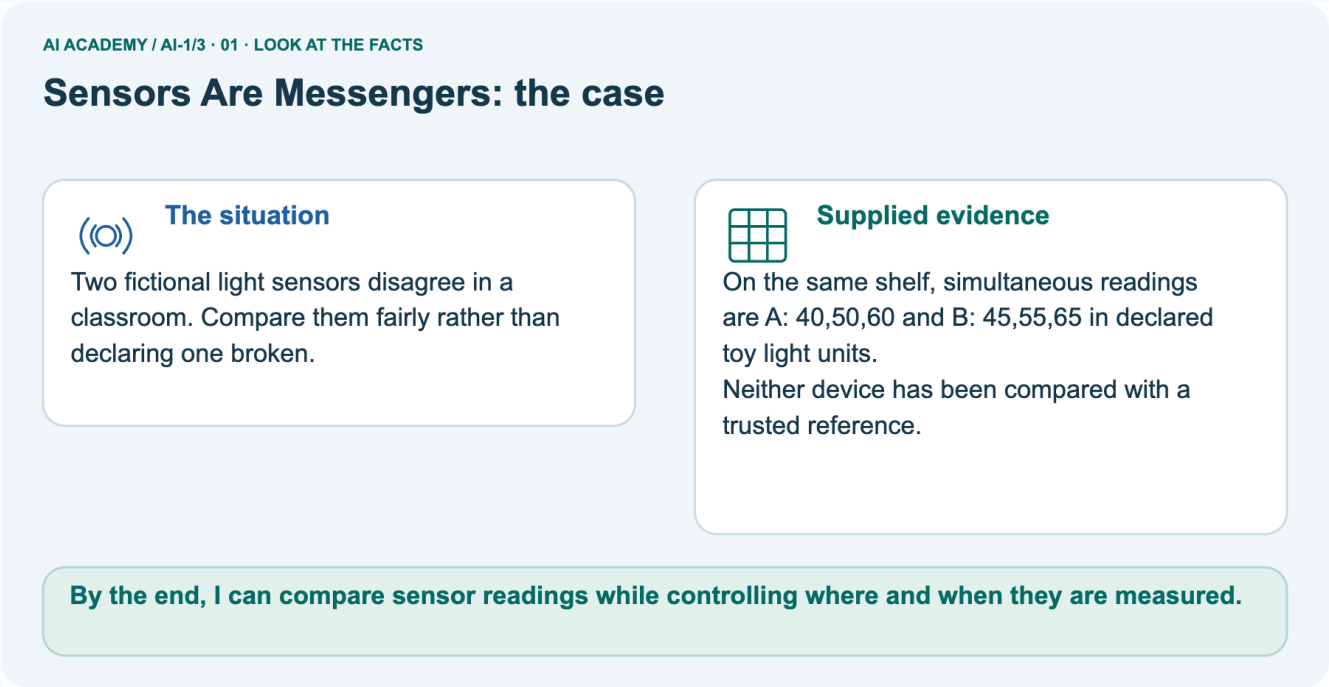
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose a useful image; completing all six again is optional.

01. Sensors Are Messengers: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

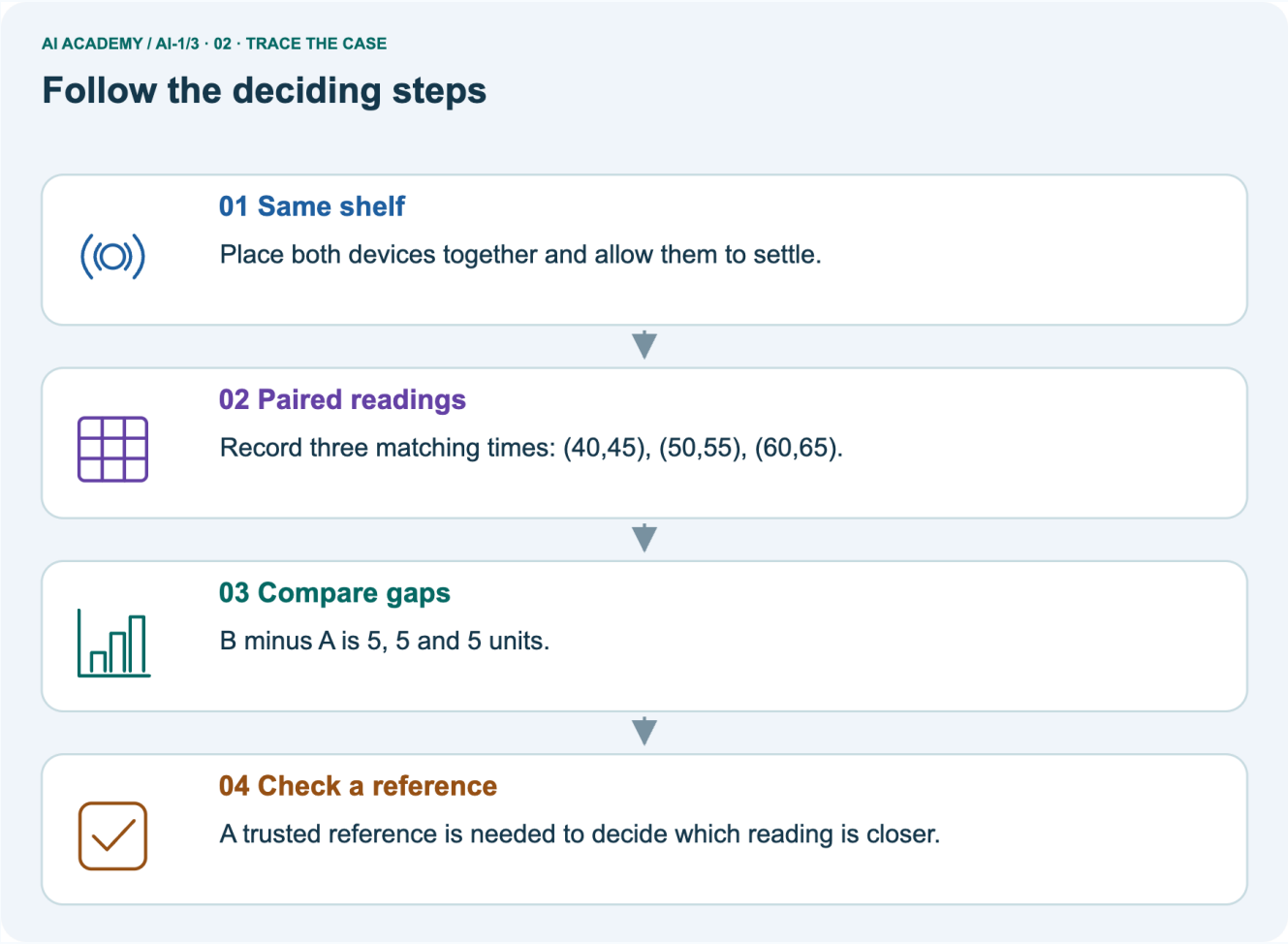
Two fictional light **sensor**s disagree in a classroom. Compare them fairly rather than declaring one broken. Supplied facts: On the same shelf, simultaneous readings are A: 40,50,60 and B: 45,55,65 in declared toy light **unit**s.

Neither device has been compared with a trusted reference.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



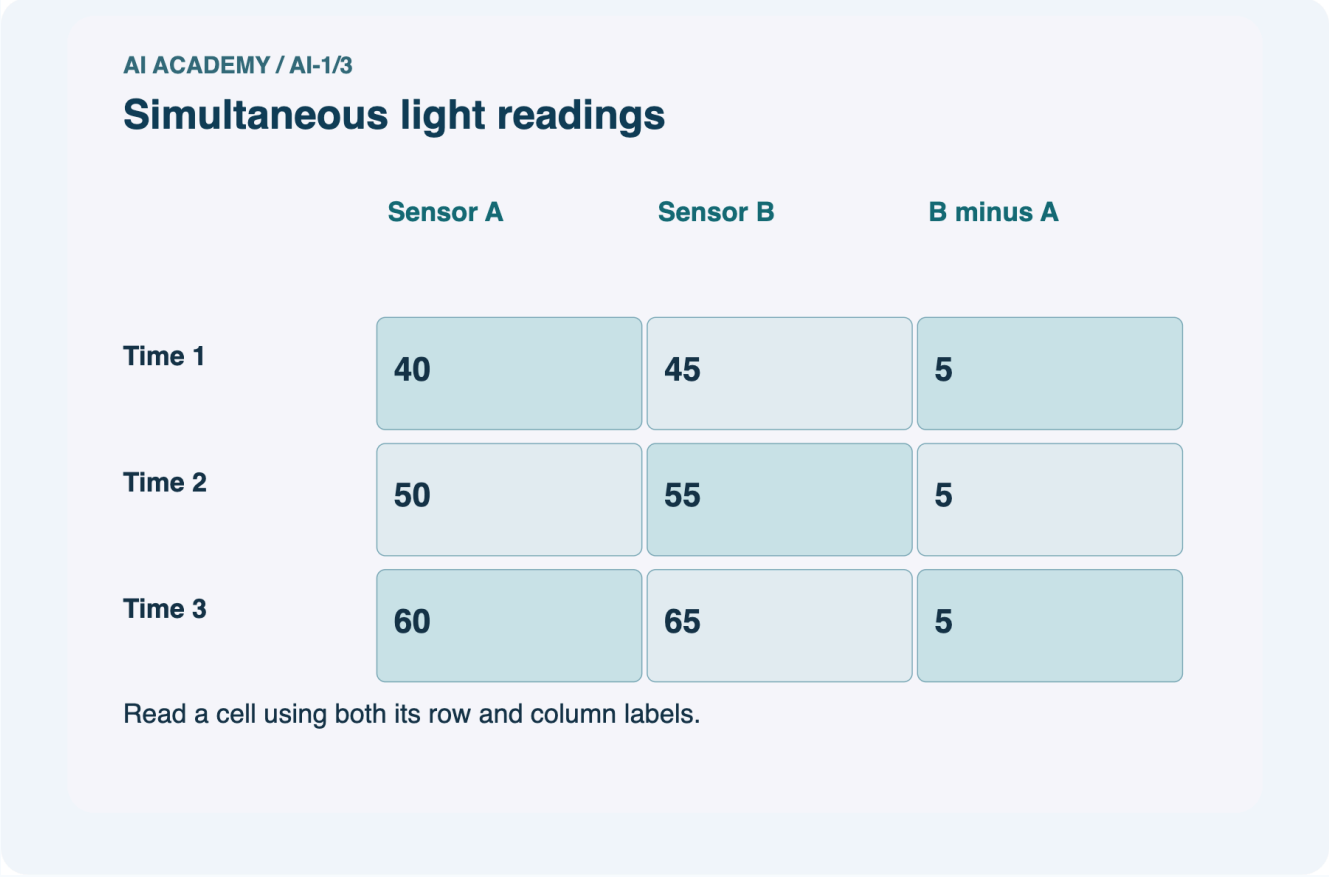
Follow the labelled picture with the matching explanation. Open the vector diagram

Same shelf: Place both devices together and allow them to settle. → Paired readings: Record three matching times: (40,45), (50,55), (60,65). → Compare gaps: B minus A is 5, 5 and 5 **unit**s. → Check a reference: A trusted reference is needed to decide which reading is closer.

Point to each numbered stage and explain which recorded **input** it uses.

> **My picture explanation, deciding evidence and one limit**

03. Simultaneous light readings



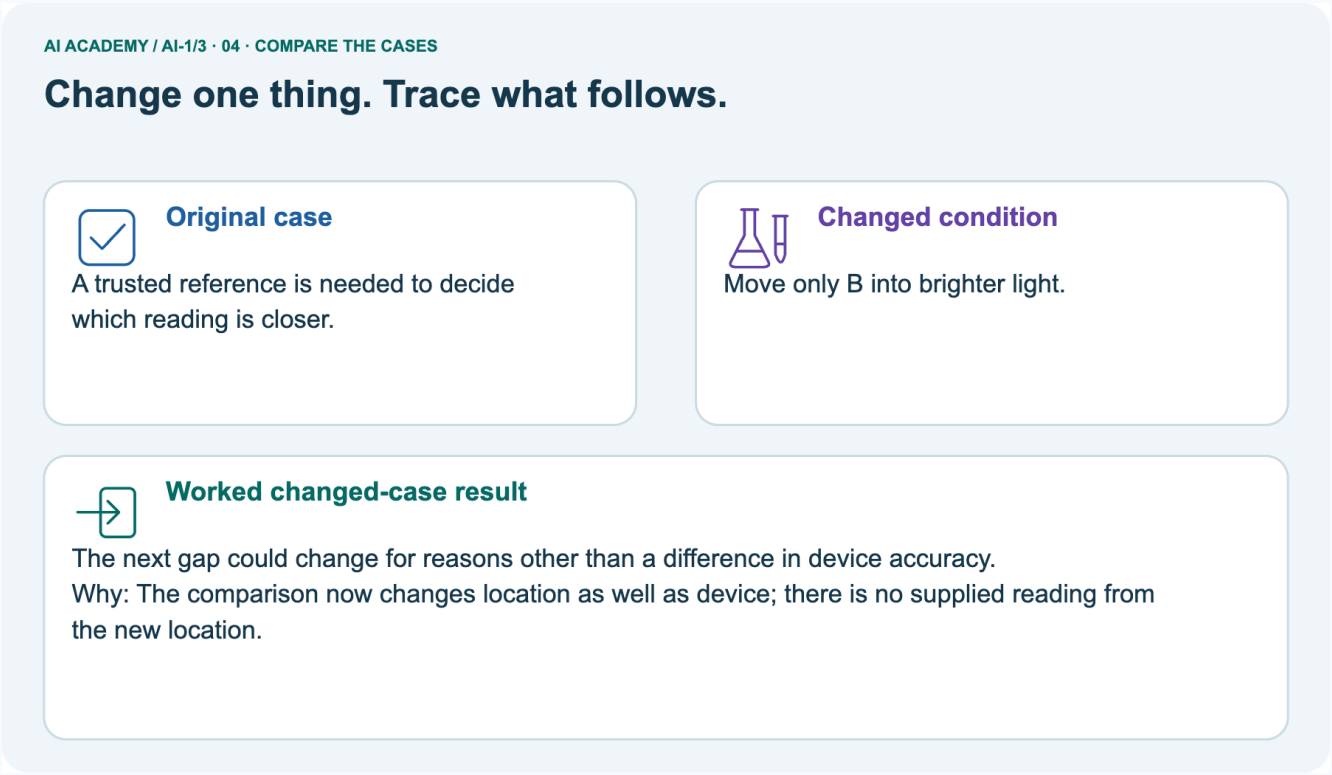
Follow the labelled picture with the matching explanation. Open the vector diagram

Simultaneous light readings. Time 1: **Sensor** A = 40, **Sensor** B = 45, B minus A = 5. Time 2: **Sensor** A = 50, **Sensor** B = 55, B minus A = 5. Time 3: **Sensor** A = 60, **Sensor** B = 65, B minus A = 5.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

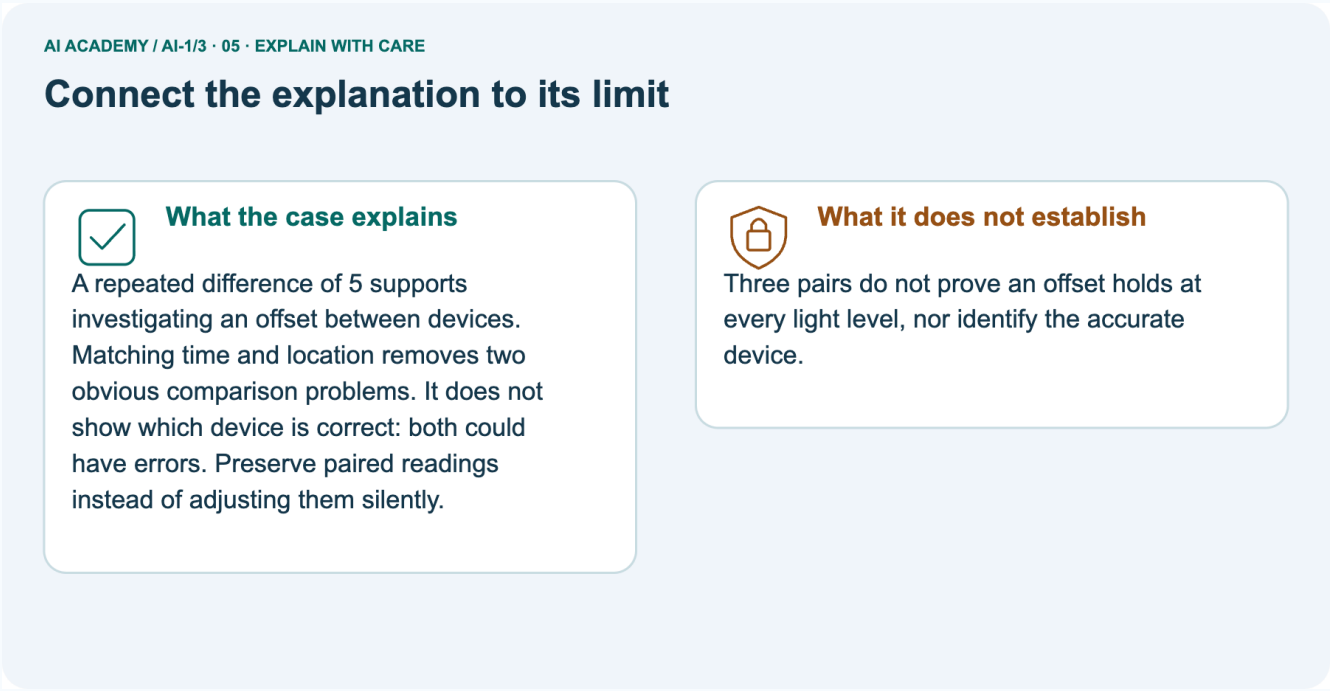
Original case: A trusted reference is needed to decide which reading is closer. Changed condition: Move only B into brighter light.

Expected result: The next gap could change for reasons other than a difference in device accuracy. Reason: The comparison now changes location as well as device; there is no supplied reading from the new location.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

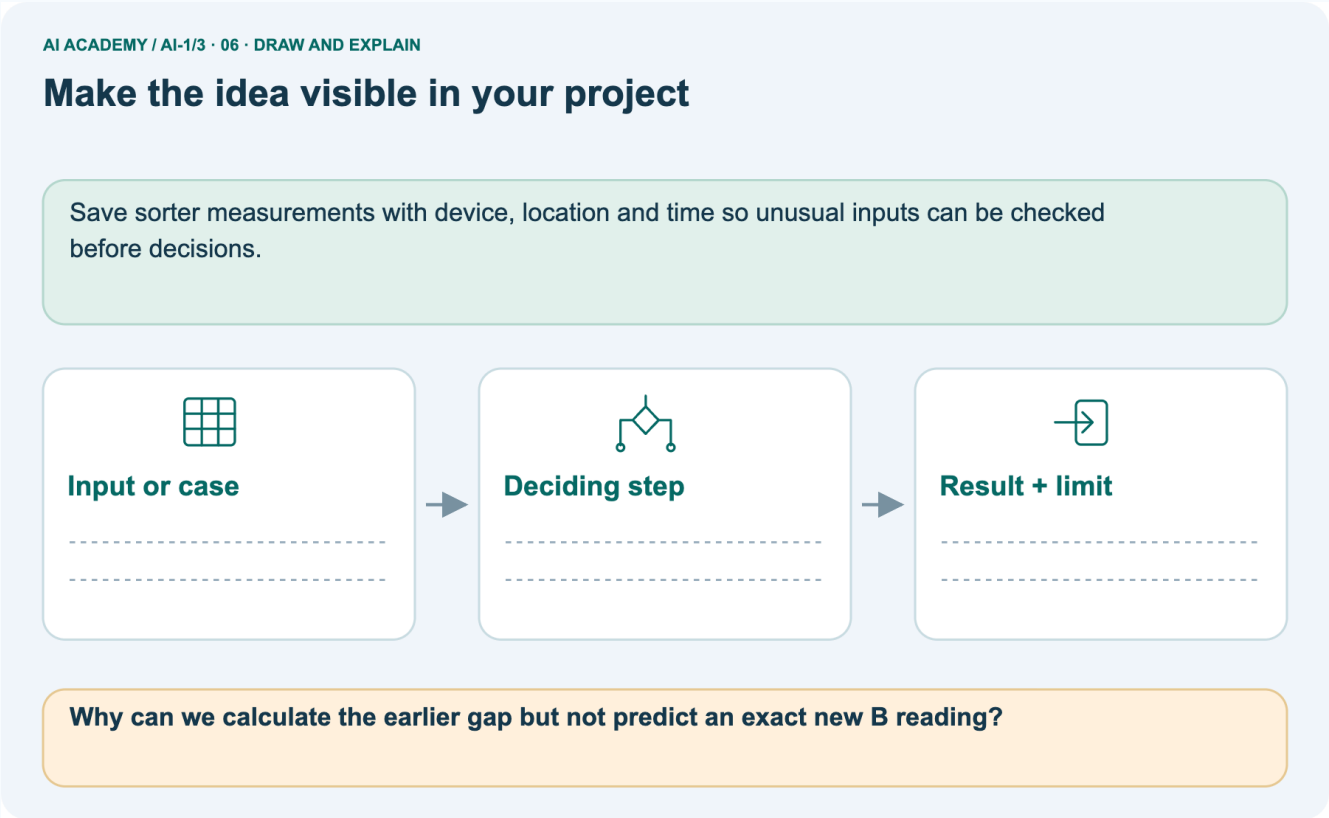
A repeated difference of 5 supports investigating an offset between devices. Matching time and location removes two obvious comparison problems. It does not show which device is correct: both could have errors.

Preserve paired readings instead of adjusting them silently. Limit: Three pairs do not prove an offset holds at every light level, nor identify the accurate device.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Save sorter **measurement**s with device, location and time so unusual **input**s can be checked before decisions. Draw your **input**, deciding step, result and limit.

Supported question: Why can we calculate the earlier gap but not predict an exact new B reading?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternative practice and picture transcripts

01. Sensors Are Messengers: the case

02. Follow the deciding steps

03. Simultaneous light readings

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.